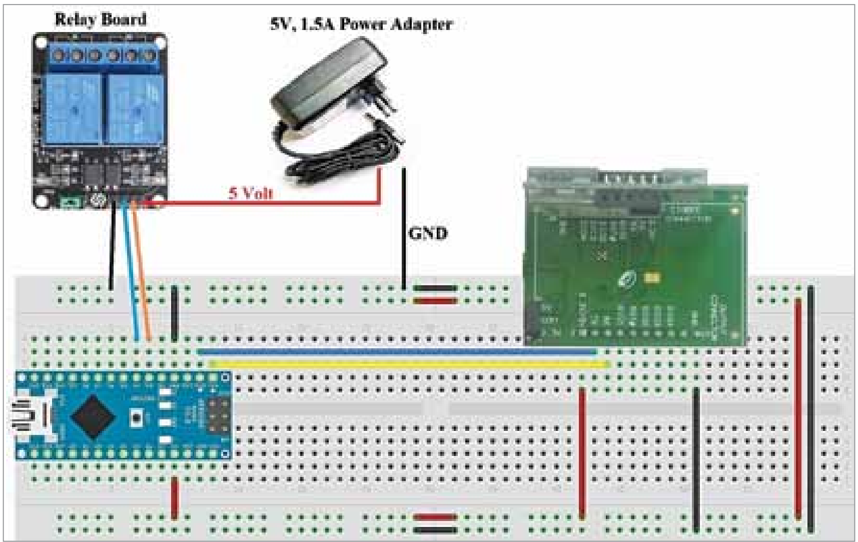


Fig. 6: Receiver circuit wired on a breadboard

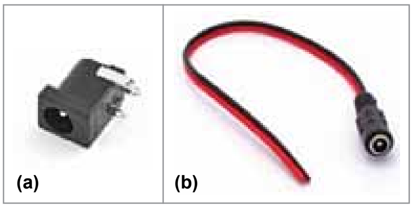


Fig. 7: (a) DC jack for PCB, (b) DC socket with wires

Fig. 8: DC jack/socket with Dupont wires soldered

8. From the PC, open Docklight for YM1 and then open a new Docklight window for YM2.

9. Check communication between the modules by sending sample data command such as 4D 08 FF FF FF 08 01 02. If YM1 receives the data from YM2 and vice versa, these devices are ready to be used for controlling appliances.

In this project, two Arduino Nano boards have been interfaced with Yoda modules for controlling two electrical appliances. Two switches (S1 and S2) are connected to digital pins D3 and D4 of Arduino Board1 in the transmitter to control two electrical loads in the receiver.

Software

Software for Arduino Nano is written in Arduino programming language, which provides serial communication via UART, SPI, and I2C. Arduino IDE is used for compiling and uploading the files to Arduino boards (Board1 and Board2).

To upload transmitter code in Arduino Nano (Board1):

1. Open Arduino IDE
2. Copy and paste transmitter code 'Remote_Tx.ino' in the Arduino IDE
3. Remove UART wires (if already connected on Rx and Tx pins) from Arduino Nano(Board1)
4. Compile the code and upload it into Arduino Nano Board1
5. Connect back Yoda UART wires to Arduino Nano Board1

To upload receiver code in Arduino Nano (Board2) follow the steps below:

1. Open Arduino IDE
2. Copy and paste the receiver code 'Relay_Rx.ino' in the Arduino IDE
3. Remove UART wires (if already connected on Rx and Tx pins) from Board2
4. Compile the code and upload it into Arduino Nano Board2
5. Connect back Yoda UART wires to Arduino Nano Board2

EFY Note
The source code of this project is available for free download at *source.efymag.com*

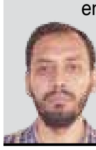
PARTS LIST	
Semiconductors:	
Board1, Board2	- Arduino Nano
YM1, YM2	- Yoda module (Zigbee)
YA1, YA2	- Yoda adaptor
Resistors (all 1/4-watt, ±5% carbon):	
R1, R2	- 10-kilo-ohm
Miscellaneous:	
RM1	- 2-channel, 5V relay module
S1, S2	- Push-to-on switch
CON1	- 2-pin connector for input DC power supply
	- Breadboard (2 nos.)
	- Hook-up wires for circuit connections in breadboards
	- Dupont jumper wires for DC jack
	- Two electrical appliances such as bulb and fan
	- 5V, 1A power adaptor

Testing

Power on the transmitter circuit first because Yoda coordinator (YM1) is responsible for starting the Zigbee communication network. Power on the receiver circuit next and wait for five seconds as defined in receiver code. Press pushbutton (S1 or S2) in transmitter circuit to control the corresponding relay in receiver circuit. Digital pins D3 and D4 of Arduino Board2 are interfaced to relay module (RM1) to switch the appliances on/off.

If on pressing switch S1 relay 1 in RM1 gets energised, it means the electrical appliance, say bulb, connected to its contacts will get turned on. Similarly, you can control relay 2 and the appliance, say fan, connected to it by pressing switch S2 from transmitter side.

The wireless range between the transmitter and the receiver depends on the Yoda modules used, which was found to be about 600 metres line-of-sight in this case. EFY



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